

SRIRAM NARAYANAN

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Research Scientist, Department of Mechanical, Materials and Aerospace Engineering, West Virginia University

EDUCATION

The Ohio State University

Ph.D. Aerospace Engineering

Columbus, OH

Jan 2021 – May 2025

- **Advisors:** Mrinal Kumar, Debdipta Goswami
- **Concentrations:** Data-driven modeling and machine learning applications on complex systems, space domain awareness, Koopman operator theory and applications

University of Southern California

Master of Science in Aerospace/Astronautical Engineering

Los Angeles, CA

July 2017 – May 2019

- **Advisors:** David Barnhart, Mike Gruntman
- **Concentrations:** Space systems engineering, satellite avionics, and optimal control

Anna University

Bachelor of Engineering in Aeronautical/Aerospace Engineering

Chennai, India

July 2013 – April 2017

RESEARCH INTERESTS

I work on the estimation and control of dynamical systems using data-driven system identification and machine-learning methods. My main areas of interest are:

- **Estimation and control of partially observable systems:** Learning dynamics from partial, low-fidelity, or indirect measurements; data-driven modeling and control of complex physical systems.
- **Robotics and control:** Machine-learning methods for robot autonomy and data-driven control in partially observable environments.
- **Space domain awareness:** Orbit determination, propagation, and predictive modeling for tracking; uncertainty propagation and data association, from LEO to cislunar space.
- **Reduced-order space weather modeling:** Fast surrogate models and data assimilation for thermospheric density and atmospheric drag.

PUBLICATIONS

18. S. Rajkumar, **S. Narayanan**, and D. Goswami, “K-PWM: Control-oriented structured world models under partial observation,” in *Advances in Neural Information Processing Systems (NeurIPS 2026)*, 2026. Accepted, to appear
17. **S. Narayanan**, D. Sicoli, H. W. Lee, and P. Mehta, “Observation-driven adaptation of reduced-order thermospheric density models,” in *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*, 2026
16. G. J. Rivera Santos, **S. Narayanan**, and P. Mehta, “SPCE-KF: Uncertainty quantification for orbit propagation using Kalman filtering and stochastic polynomial chaos expansion,” in *AAS/AIAA Astrodynamics Specialist Conference 2026*, 2026. Paper AAS 26-960
15. C. Lashley, R. A. Appana, **S. Narayanan**, S. Chakravorty, and M. Kumar, “Small eccentricity restricted three body problem,” in *AAS/AIAA Astrodynamics Specialist Conference 2026*, 2026. Paper AAS 26-972

14. **S. Narayanan**, M. N. Gul Mohamed, I. Paranjape, I. Nayak, S. Chakravorty, and M. Kumar, "State forecasting in an estimation framework with surrogate sensor modeling," in *AAS/AIAA Astrodynamics Specialist Conference 2025*, 2025. Paper AAS 25-577
13. **S. Narayanan**, M. N. Gul Mohamed, I. Nayak, S. Chakravorty, and M. Kumar, "On the predictive capability of dynamic mode decomposition for periodic solutions of nonlinear systems with focus on orbital mechanics," *The Journal of the Astronautical Sciences*, vol. 72, no. 3, p. 23, 2025
12. **S. Narayanan**, M. N. Gul Mohamed, I. Nayak, S. Chakravorty, and M. Kumar, "Time-delayed dynamic mode decomposition for families of periodic trajectories in cislunar space," in *AAS/AIAA Astrodynamics Specialist Conference 2024*, 2024. Paper AAS 24-293
11. **S. Narayanan**, M. N. Gul Mohamed, I. Nayak, S. Chakravorty, and M. Kumar, "On dynamic mode decomposition for nonlinear periodic systems with a focus on orbital mechanics," in *AAS/AIAA Astrodynamics Specialist Conference 2024*, 2024. Paper AAS 24-121
10. M. Wolf, R. Balo, **S. Narayanan**, and M. Kumar, "On solar radiation pressure: Influence of area-to-mass ratio uncertainty and various shadow models," in *AAS/AIAA Astrodynamics Specialist Conference 2024*, 2024. Paper AAS 24-134
9. D. Maity, D. Goswami, and **S. Narayanan**, "On the effect of quantization on dynamic mode decomposition," *IEEE Conference on Decision and Control*, 2024
8. A. Cortez, B. Ford, I. Nayak, **S. Narayanan**, and M. Kumar, "Hybrid A* path search with resource constraints and dynamic obstacles," *Frontiers in Aerospace Engineering*, vol. 1, p. 1076271, 2023
7. A. C. Cortez, B. T. Ford, I. Nayak, **S. Narayanan**, and M. Kumar, "Path planning for a Dubins agent with resource constraints and dynamic obstacles," in *AIAA SCITECH 2023 Forum*, p. 1054, 2023
6. **S. Narayanan**, I. Nayak, and M. Kumar, "An iterative scheme to learn system dynamics of space objects from partial state information," in *AIAA SCITECH 2023 Forum*, p. 0124, 2023
5. **S. Narayanan**, I. Nayak, and M. Kumar, "On the application of time delay embedding for the data driven discovery of nonlinear systems from partial state information," in *AIAA SCITECH 2022 Forum*, p. 2440, 2022
4. G. Minster, A. Chang, J. B. Inouye, **S. Narayanan**, A. Carter, J. Tong, and D. A. Barnhart, "Hyperion: Artificial gravity reusable crewed deep space transport," *Journal of Space Safety Engineering*, vol. 7, no. 1, pp. 3–10, 2020
3. **S. Narayanan**, D. Barnhart, R. Rogers, D. Ruffatto, E. Schaler, N. Van Crey, G. Dean, A. Bhanji, S. Bernstein, A. Singh, *et al.*, "REACCH-reactive electro-adhesive capture cloth mechanism to enable safe grapple of cooperative/non-cooperative space debris," in *AIAA Scitech 2020 Forum*, p. 2134, 2020
2. R. Rughani, R. Rogers, J. Allam, **S. Narayanan**, P. Patil, K. Clarke, M. Lariviere, J. Plessis, L. Na, D. Healy, *et al.*, "Improved CubeSat mission reliability using a rigorous top-down systems-level approach," in *70th International Astronautical Congress*, 2019
1. **S. Narayanan**, R. Rughani, R. Rogers, K. Clarke, and J. Allam, "Mission Dodona: Electronic power system design, analysis and integration," in *Small Satellite Conference*, 2019

UNDER REVIEW

3. D. Sicoli, **S. Narayanan**, and P. Mehta, "Fast and stable nonlinear emulation of TIEGCM using an autoregressive SINDy framework," *AGU Space Weather*, 2026. Submitted

2. **S. Narayanan**, D. Sicoli, and P. Mehta, “Reduced-order data assimilation for thermospheric density using physics-informed SINDyc models,” *AGU Space Weather*, 2026. Submitted
1. **S. Narayanan**, G. J. Rivera Santos, Z. Toler, B. A. Jones, and P. Mehta, “Stochastic polynomial chaos Kalman filtering for cislunar orbit determination under model uncertainty,” in *37th AAS/AIAA Space Flight Mechanics Meeting, New Orleans, LA*, 2027. Extended abstract submitted

IN PREPARATION

1. **S. Narayanan**, H. W. Lee, and P. Mehta, “Optimal observing systems for reduced-order thermospheric density estimation,” *Acta Astronautica*, 2026. In preparation

PROFESSIONAL EXPERIENCE

West Virginia University

Research Scientist

Morgantown, WV

June 2025 – Present

- Applying machine learning, data assimilation, and filtering techniques to improve space weather forecasting through better integration of observational data and increased model accuracy.
- Developing an optimal observing-system design framework that selects which orbits to observe from to maximize the information delivered to reduced-order density estimators.
- Contributing preliminary results to externally sponsored proposals now under review, including with NOAA.
- Mentoring undergraduate and graduate students’ research.

The Ohio State University

Graduate Research/Teaching Associate - Ph.D. Candidate

Columbus, OH

Dec 2020 – May 2025

- Improving machine learning tools to track objects in space better, using advanced state prediction methods even with limited data, and applying these techniques in robotics and beyond.
- Implementing data-driven state estimators and filters for Space Domain Awareness (SDA) applications, grounded in control theory principles for reliable estimation under uncertainty.
- Mentoring students on research activities, emphasizing control theory, state estimation, and aerospace applications.
- Teaching mechanical and aerospace engineering courses, including control theory, state estimation, and dynamic system modeling, focusing on robotics applications and autonomous systems.

PointPro USA

Research Engineering Intern

Columbus, OH

May 2024 – Aug 2024

- Supported PointPro’s backend software operations at the Space Systems Command’s (SSC), SDA Tools Applications and Processing (TAP) Lab.
- Improved an adaptive Monte Carlo detection and threat estimation framework to generate more confident results in a highly uncertain environment.
- Developed and tested an optimal measurement update algorithm to generate new uncertainty point clouds for threat assessment.
- Built visualization tools and data packages for downstream delivery and integration with other cohort members in the Apollo Accelerator program.

Panasonic Avionics Corporation

Systems Engineer

Lake Forest, CA

Feb 2020 – Nov 2020

- Built and integrated flight infotainment systems and tested engineering kits for delivery purposes.
- Tested, troubleshot, and triaged issues to validate and certify products before delivery.
- Maintained laboratory test racks that house all systems for several aircraft and customers.
- Created, planned, and managed system requirements, drafted compliance, and test procedures per FAA regulatory standards.

Information Sciences Institute

Graduate Research Assistant/Research Engineer

Marina Del Rey, CA

May 2018 – May 2019 / June 2019 – Jan 2020

- Led power sub-system development for Dodona (La Jument 2) satellite → Created test plans, built test equipment, installed DAQ systems using programmable laboratory instruments, and created procedures to maintain compliance.
- Performed comprehensive evaluation of electronics such as solar panels and battery systems through standalone tests, in-circuit tests, and depth of discharge analysis.
- Worked in collaboration with NASA-Jet Propulsion Laboratory to develop a gecko-adhesion-based robot for on-orbit servicing missions → Created a control/physics simulation to characterize sensor behavior and system performance, simulation tools used: MATLAB Simulink, Project Chrono, and Gazebo.
- Supported USC undergraduate students in the development of MAGNETO satellite.

University of Southern California

Graduate Research Assistant

Los Angeles, CA

Dec 2017 – May 2018

- Worked in trajectory optimization for spacecraft: applied optimal control theory to optimize a low thrust rendezvous trajectory.
- Developed an optimizing routine for a two-body rendezvous scenario.

Flight Laboratory, Indian Institute of Technology

Trainee

Kanpur, India

Dec 2016 – Jan 2017

- Performed weight and flight performance estimation studies on Cessna 206H Stationair and NAL Hansa-3 aircraft.
- Carried out calibration of control surfaces for NAL Hansa-3 aircraft.

TEACHING EXPERIENCE

- **AE 4510/11: Experimental Projects (Senior Capstone):** Graduate Teaching Associate (SP2023, SP2024, AU2024). Supervised senior design teams through the full project lifecycle (team formation, requirements, design reviews, build, and test), and steered projects toward space domain awareness applications where team interests allowed.
- **AE 2200: Introduction to Aerospace Engineering I:** Graduate Teaching Associate (AU2023). Ran an independent lab section with full autonomy over classroom teaching and student interaction for first-year aerospace students.
- **AE 3520: Flight Vehicle Dynamics:** Graduate Teaching Associate (SP2023, SP2024). Wrote and graded assignments, held office hours, and relayed student difficulties back to the instructor of record.
- **AE 3581: Numerical Methods in Aerospace Engineering:** Graduate Teaching Associate (AU2023).
- **AE 3542: Flight Vehicle Structures I:** Graduate Teaching Associate (AU2024).
- **Teaching development:** Completed Ohio State's Future Academic Scholars Training (FAST) program and Drake Institute teaching seminars.

MENTORSHIP

- **Daniele Sicoli:** Ph.D. Student, Aerospace Engineering, WVU (2025 – Present)
- **Gerardo Rivera Santos:** Ph.D. Student, Aerospace Engineering, WVU (2025 – Present)
- **Abhishek Appana:** Ph.D. Student, Aerospace Engineering, OSU (2025 – Present)
- **Connor Lashley:** Ph.D. Student, Aerospace Engineering, OSU (2025 – Present)
- **Santhosh Rajkumar:** Ph.D. Student, Mechanical Engineering, OSU (2024 – Present)
- **Mark Wolf:** M.S. Student, Aerospace Engineering, OSU (2024 – 2025)
- **Ian Binder:** Undergraduate Student, Aerospace Engineering, OSU (2022 – 2023)

INVITED TALKS AND POSTERS

- **UCAR Space Weather Workshop:** Bridging High-Fidelity Modeling and Operations with Reduced-Order Thermospheric Data Assimilation (April 2026)
- **American Control Conference (ACC) Workshop: Recent Developments in Data-Driven Methods for Dynamical Systems and Control:** Iterative Learning of Dynamics of Physical Systems from Partial State Information (May 2023)
- **Edward F. Hayes Advanced Research Forum (2023):** Poster session: Data-driven identification and forecasting of space object dynamics from partial measurements (February 2023)

SERVICE

- **Session Chair at AIAA SciTech:** Learning, Reasoning, and Data-Driven Systems (Jan 2023)
- **Faculty Hiring Committee, Mechanical and Aerospace Engineering, OSU:** Student member
- **Reviewer:** AIAA - Journal of Aerospace Information Systems
Elsevier - Aerospace Science and Technology
Elsevier - Acta Astronautica
Springer - The Journal of the Astronautical Sciences
Modeling, Estimation and Control Conference (MECC) - 2024 (x2)

MEMBERSHIPS

- **AIAA - American Institute of Aeronautics and Astronautics:** Member
- **IEEE - Institute of Electrical and Electronics Engineers:** Member
- **AAS - American Astronautical Society:** Member

KEY SKILLS

Software:

- MATLAB (including Simulink), Python, C++, C – Proficient in scientific software development.
- Unix – Knowledge of scripting, system administration and experience developing on embedded platforms.
- LaTeX, Microsoft Office, Git – Proficient.
- AGI Satellite Toolkit – Level 1 Certification.
- SOLIDWORKS, ANSYS, AutoCAD – Designing and simulation (working knowledge).

Scientific Computing and Engineering Tools:

- Prior experience with NumPy, SciPy, Pandas, Matplotlib, PyTorch, TensorFlow, Numba, CVXPY, Eigen, Boost, Ceres Solver, CasADi, ACADOS, OptiToolbox, Ipopt, MATLAB Optimization Toolbox, SNOPT
- Robotics and Control: ROS (Robot Operating System), Gazebo

REFERENCES

- **Piyush Mehta (piyush.mehta@mail.wvu.edu):** Associate Professor, Mechanical, Materials and Aerospace Engineering, WVU
- **Mrinal Kumar (kumar.672@osu.edu):** Professor, Mechanical and Aerospace Engineering, OSU
- **Brian Ritchie (ritchie.11@osu.edu):** Professional Practice Assistant Professor, Mechanical and Aerospace Engineering, OSU; instructor of record for my capstone teaching
- **Debdipta Goswami (goswami.78@osu.edu):** Assistant Professor, Mechanical and Aerospace Engineering, Electrical and Computer Engineering, OSU